

TECHNICAL PROJECT REPORT

**PV SYSTEM WITH INCREMENTAL CONDUCTANCE
MPPT
AND GRID INTEGRATION**

MATLAB/Simulink – Simscape Electrical

Detailed Technical Documentation | Model: sgs_assin_2.slx

Abstract

This project develops a grid-connected photovoltaic conversion system in MATLAB/Simulink using Incremental Conductance MPPT, DC-link PI control, a single-phase full-bridge VSC, PLL synchronization, PR current control and PWM switching. The test scenario changes irradiance from 500 to 1000 W/m² at 0.5 s while temperature remains at 25 °C.

Updated System Model

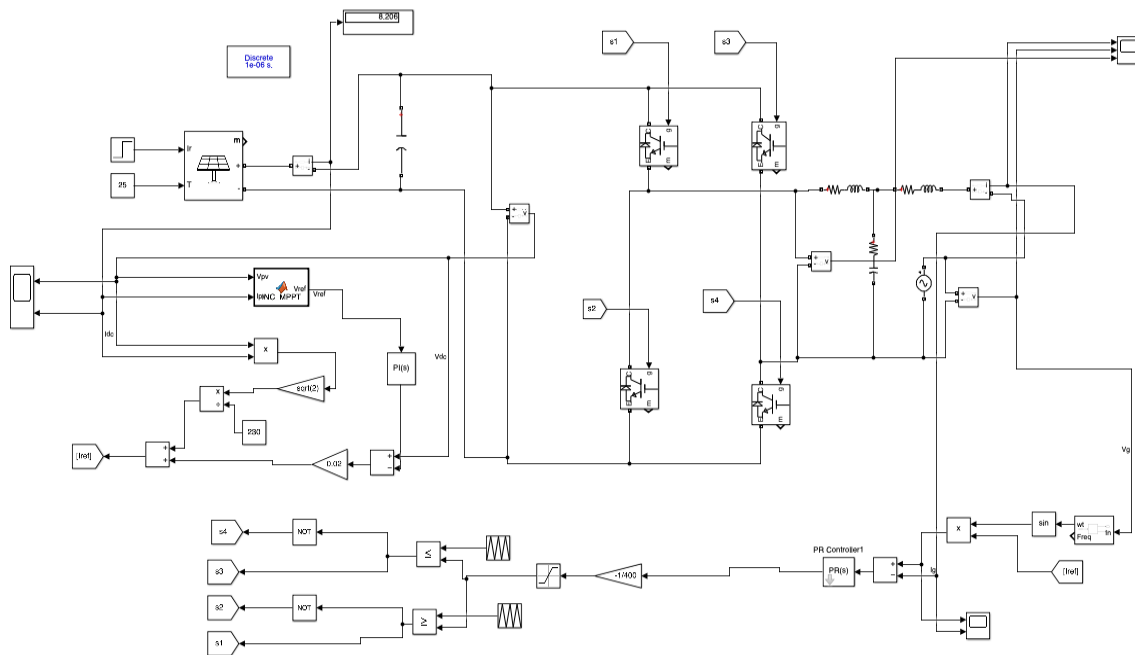


Figure 1. Updated complete MATLAB/Simulink model.

1. Introduction

The model combines PV generation, maximum-power extraction, DC-link regulation, voltage-source conversion, grid synchronization and sinusoidal current control.

2. Objectives

The project objectives are to implement INC MPPT, regulate the DC link, operate the full-bridge VSC, synchronize to a 50 Hz grid, regulate grid current with a PR controller and study the response to irradiance changes.

3. PV Array Model

The array contains 12 series modules and one parallel string. Each module is 235.008 W with $V_{mp} = 30.6$ V, $I_{mp} = 7.68$ A, $V_{oc} = 36.72$ V and $I_{sc} = 8.23$ A. Derived array V_{mp} is 367.2 V, V_{oc} is 440.64 V and nominal maximum power is approximately 2.82 kW.

4. PV Maximum-Power Principle

PV power is $P = VI$. At the maximum-power point, $dP/dV = 0$, giving $dI/dV = -I/V$. The operating point is therefore identified by comparing incremental and instantaneous conductance.

5. Incremental Conductance MPPT

The MATLAB Function receives V_{pv} and I_{pv} , stores previous values, calculates ΔV and ΔI , compares $\Delta I/\Delta V$ with $-I/V$ and updates V_{ref} . The model uses $\Delta V_{ref} = 0.0002$ and tolerance $1e-6$.

6. DC-Link Regulation

The DC link uses a 1000 μF capacitor with an initial voltage of 500 V. A PI controller with $K_p = 0.3$ and $K_i = 0.1$ regulates the DC-link voltage toward the MPPT reference. The initial capacitor energy is 125 J.

7. Full-Bridge VSC

Four IGBT/Diode devices form the single-phase full bridge. Carrier-based PWM uses a period of approximately 200 μs , corresponding to about 5 kHz.

8. AC Interface

The converter is connected to the grid through passive interface/filter elements. Their impedance affects switching ripple, current dynamics and voltage drop.

9. Grid Source and PLL

The grid is 325 V peak, approximately 230 V RMS, at 50 Hz. The PLL estimates phase and frequency for synchronization.

10. PR Current Control

The PR controller uses $k_p = 1.2$, $k_r = 750$ and a resonant frequency of 50 Hz. Its resonant branch increases gain around the fundamental and improves sinusoidal current tracking.

11. Control Interaction

MPPT establishes the desired PV operating voltage; the DC-link controller manages energy balance; the grid-current loop controls the AC-side current. These loops must be coordinated because changes in PV power propagate through the DC link to the grid.

12. Simulation Setup

Powergui is discrete with a 1 μ s sample time. The simulation runs for 2 s. Irradiance changes from 500 to 1000 W/m² at 0.5 s and temperature remains 25 °C.

13. Expected Response

The irradiance step should cause a change in available PV current and power. INC MPPT should move the operating point, the DC-link loop should restore voltage, and the grid-current controller should adjust power transfer while remaining synchronized.

14. Results Evaluation

Recommended plots are PV voltage/current/power, V_{ref} versus V_{pv} , DC-link voltage, grid voltage/current, I_{ref} versus I_g , active/reactive power and FFT/THD.

15. Verification Note

The physical source, PLL and PR controller use 50 Hz, while the supplied powergui configuration stores 60 Hz as a fundamental-frequency value. This should be reconciled before formal RMS/FFT reporting.

16. Engineering Interpretation

The updated model diagram clearly separates the upper power circuit from the lower control and PWM circuits. This makes signal tracing, debugging and documentation more systematic.

17. Recommended Test Matrix

In addition to the 500→1000 W/m² step, evaluate low irradiance, temperature variation and a 1000→500 W/m² negative step.

18. Troubleshooting

If V_{ref} moves in one direction, inspect INC sign conventions. If V_{dc} diverges, inspect PI polarity, modulation and power balance. If current is distorted, inspect PWM, PR frequency and filter parameters.

19. Reproducibility

Use the supplied model, verify all listed parameters, run for 2 s, export the final waveforms and calculate steady-state metrics over a stated analysis window.

20. GitHub and Publication

Keep the model, controller code, documentation and figures organized. Check course/institutional rules before making a laboratory .slx model public.

21. Conclusion

The model provides an integrated study platform for PV MPPT, DC-link dynamics, VSC switching, grid synchronization and current control. Quantitative performance should be added from final exported simulation data.

Appendix – Key Parameters

Subsystem	Parameter	Value
PV Array	Nser / Npar	12 / 1
PV Array	Vmp / Imp per module	30.6 V / 7.68 A
PV Array	Voc / Isc per module	36.72 V / 8.23 A
MPPT	ΔV_{ref} / tolerance	0.0002 / 1e-6
DC Link	C / initial V	1000 μ F / 500 V
PI	Kp / Ki	0.3 / 0.1
VSC	Topology	Single-phase full bridge
PWM	Frequency	$\approx$ 5 kHz
Grid	Peak voltage / frequency	325 V / 50 Hz
PR	kp / kr / f	1.2 / 750 / 50 Hz
Powergui	Mode / Ts	Discrete / 1 μ s
Test	Irradiance step	500 $\rightarrow$ 1000 W/m ² at 0.5 s